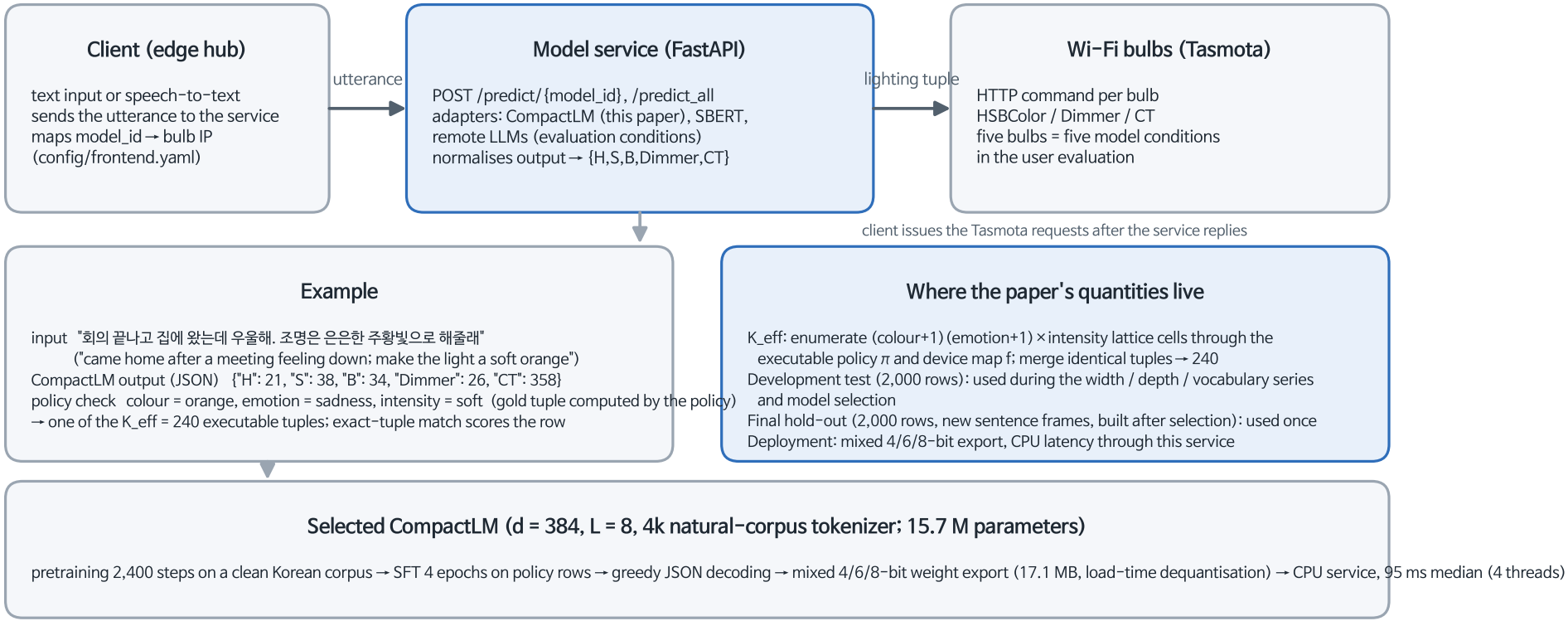

Client (edge hub)
text input or speech-to-text
sends the utterance to the service
maps model_id → bulb IP
(config/frontend.yaml)
utterance
Model service (FastAPI)
POST /predict/{model_id}, /predict_all
adapters: CompactLM (this paper), SBERT,
remote LLMs (evaluation conditions)
normalises output → {H,S,B,Dimmer,CT}
lighting tuple
Wi-Fi bulbs (Tasmota)
HTTP command per bulb
HSBColor / Dimmer / CT
five bulbs = five model conditions
in the user evaluation
client issues the Tasmota requests after the service replies
Example
input "회의 끝나고 집에 왔는데 우울해. 조명은 은은한 주황빛으로 해줄래"
("came home after a meeting feeling down; make the light a soft orange")
CompactLM output (JSON) {"H": 21, "S": 38, "B": 34, "Dimmer": 26, "CT": 358}
policy check colour = orange, emotion = sadness, intensity = soft (gold tuple computed by the policy)
→ one of the K_eff = 240 executable tuples; exact-tuple match scores the row
Where the paper's quantities live
K_eff: enumerate (colour+1) (emotion+1) × intensity lattice cells through the
executable policy π and device map f; merge identical tuples → 240
Development test (2,000 rows): used during the width / depth / vocabulary series
and model selection
Final hold-out (2,000 rows, new sentence frames, built after selection): used once
Deployment: mixed 4/6/8-bit export, CPU latency through this service
Selected CompactLM (d = 384, L = 8, 4k natural-corpus tokenizer; 15.7 M parameters)
pretraining 2,400 steps on a clean Korean corpus → SFT 4 epochs on policy rows → greedy JSON decoding → mixed 4/6/8-bit weight export (17.1 MB, load-time dequantisation) → CPU service, 95 ms median (4 threads)